

## **SET COVERING ALGORITHMS USING CUTTING PLANES, HEURISTICS, AND SUBGRADIENT OPTIMIZATION: A COMPUTATIONAL STUDY**

Egon BALAS\* and Andrew HO

*Carnegie-Mellon University, Pittsburgh, PA, U.S.A.*

Received 30 July 1979

Revised manuscript received 18 October 1979

We report on the implementation and computational testing of several versions of a set covering algorithm, based on the family of cutting planes from conditional bounds discussed in the companion paper [2]. The algorithm uses a set of heuristics to find prime covers, another set of heuristics to find feasible solutions to the dual linear program which are needed to generate cuts, and subgradient optimization to find lower bounds. It also uses implicit enumeration with some new branching rules. Each of the ingredients was implemented and tested in several versions. The variant of the algorithm that emerged as best was run on 55 randomly generated test problems (20 of them from the literature), with up to 200 constraints and 2000 variables. The results show the algorithm to be more reliable and efficient than earlier procedures on large, sparse set covering problems.

*Key words:* Set Covering Problem, Cutting-planes, Subgradient Optimization, Computation, Heuristics, Algorithms, Branch and Bound.

### **1. Introduction: Cutting planes from conditional bounds**

In this paper we report on the implementation and computational testing of an algorithm, or rather a class of algorithms, based on the cutting planes from conditional bounds introduced in [1], and also using as ingredients several heuristics for finding feasible primal and dual solutions, subgradient optimization of a Lagrangean function, and implicit enumeration with some new branching rules.

The family of cutting planes from conditional bounds is briefly described in this section; a more detailed discussion of the properties of these cuts is to be found in the companion paper [2]. In Section 2 we describe the general features of the algorithm whose versions we implemented and tested. Sections 3–7 discuss each of the ingredients of our procedure, with comparisons of various versions based on computational testing. Finally, Section 8 summarizes our computational experience with the algorithm as a whole. While the discussion focuses on a particular instance of the algorithms in the class considered, namely the one that emerged as the most successful from our computational experiments, we also discuss possible variations wherever it seems useful.

\* Research supported by the National Science Foundation under grant MCS76-12026 A02 and the Office of Naval Research under contract N00014-75-C-0621 NR 047-048.

As one can see from the computational results presented in Section 8, the algorithm discussed here is a reliable and efficient tool for solving large, sparse set covering problems of the kind that frequently occur in practice. With a time limit of 10 minutes on the DEC 20/50, we have solved all but one of a set of 50 randomly generated set covering problems with up to 200 constraints, 2000 variables, and 8000 nonzero matrix entries (here "solving" means finding an optimal solution and proving its optimality), never generating a branch and bound tree with more than 50 nodes. For problems that are too large to be solved within a reasonable time limit, the procedure usually finds good feasible solutions, with a bound on the distance from the optimum (for the one unsolved problem, this bound was 2.3%). We also tested the algorithm on a set of 5% density problems, but as density increases, the performance of the algorithm tends to decline.

We consider the set covering problem

$$(SC) \quad \min\{cx \mid Ax \geq e, x \in \{0, 1\}^n\}$$

where  $A = (a_{ij})$  is an  $m \times n$  0-1 matrix, and  $e = (1, \dots, 1)$  has  $m$  components. Let  $a^i$  and  $a_j$  denote the  $i$ -th row and the  $j$ -th column of  $A$ , and let  $M = \{1, \dots, m\}$ ,  $N = \{1, \dots, n\}$ . We denote

$$M_j = \{i \in M \mid a_{ij} = 1\}, \quad j \in N; \quad N_i = \{j \in N \mid a_{ij} = 1\}, \quad i \in M.$$

We will also use the pair of dual linear programs

$$(L) \quad \min\{cx \mid Ax \geq e, x \geq 0\}$$

and

$$(D) \quad \max\{ue \mid uA \leq c, u \leq 0\},$$

associated with (SC).

A vector  $x \in \{0, 1\}^n$  satisfying  $Ax \geq e$  is called a *cover*, and  $S(x) = \{j \in N \mid x_j = 1\}$  its *support*. A cover whose support is minimal, is *prime*. For a cover  $x$ , we denote  $T(x) = \{i \in M \mid a^i x = 1\}$ .

The theory underlying the family of cutting planes from conditional bounds can be summarized as follows (for proofs of these statements and further elaboration see the companion paper [2]).

Let  $z_U$  be some known upper bound on the value of (SC), and let  $u$  be any feasible solution to (D), with  $s = c - uA$ , such that the condition

$$\sum_{j \in S} s_j \geq z_U - ue \tag{1}$$

is satisfied for some  $S \subseteq N$ . Let  $S = \{j(1), \dots, j(p)\}$ , and let  $Q_i$ ,  $i = 1, \dots, p$ , be any collection of subsets of  $N$  satisfying

$$\sum_{i \mid j \in Q_i} s_{j(i)} \leq s_j, \quad j \in N. \tag{2}$$